Fiscal Year:	FY 2005	**Task Last Updated:**	FY 05/26/2005
PI Name:	Fitts, Robert H Ph.D.		
Project Title:	Effect of Prolonged Space Flight on Human Skeletal Muscle		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Physiology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Milwaukee	**State:**	WI
Zip Code:	53201	**Congressional District:**	4
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	96-HEDS-04
Start Date:	10/01/1998	**End Date:**	12/31/2007
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
Contact Monitor:	McCollum, Suzanne	**Contact Phone:**	281 483-7307
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Flight Program:	ISS		
Flight Assignment:	ISS Increments 4-10 In flight development phase (data collection has begun)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Trappe, Scott (Ball State University) Riley, Danny (Medical College of Wisconsin)		
Grant/Contract No.:	NCC9-116		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Our long term goals are to understand the effects of zero g space flight on the neuromuscular system, and work to develop effective exercise countermeasures. The specific goals of this project are to establish a time course for the structural and functional alterations in limb skeletal muscle with prolonged weightlessness, and from these data calculate the new microgravity steady state that would likely be reached in a trip to Mars. Since we have already made a detailed characterization of how human limb muscles react to short duration (17 days) flight, we propose to study the effects of 3 flights of progressively longer duration. The ideal design would be to study 30, 90, 180 or 30, 60, 90 days. The ideal number of subjects for each flight would be 3 to 4, but we realize from experience that the number available for a given duration flight maybe 1 or 2. This project is designed as a pre- and post-flight study. Calf muscle function will be evaluated using a specially designed torque velocity dynamometer, and biopsies will be taken from the soleus and gastrocnemius muscles at L-45 and immediately post flight (R+0). The effect of prolonged space flight on the functional and structural properties of individual fast and slow fibers will be determined with special emphasis on fiber type specific changes in cell size, peak force and power, the maximal shortening velocity, and the cells resistance to damage. Immunohistochemical studies will be carried out to assess the extent of slow to fast isozyme shifts in myosin, and EM analyses will determine the structural relationship of the thick and thin filament, the degree of myofilament loss, and the integrity of the dystrophin-dystroglycan complex. Microbiochemical studies will establish the extent to which prolonged zero-g affects the substrate and enzyme profile of each fiber type. In conclusion, we will establish the time course for functional and structural change in limb skeletal muscle with prolonged space flight, and establish mechanistic explanations for many of the observed changes. At a minimum we will establish the relative importance of changes in the myosin and actin filament structure vs qualitative changes in the myosin protein, and the relationship of the integrity of the dystrophin-dystroglycan complex to the ability of the cell to resist stretch-induced damage.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The primary benefit will be to allow countermeasures to be developed that will prevent the microgravity-induced loss of muscle mass and function, such that, prolonged missions to Mars and beyond can be undertaken. This project will establish the time course of the loss of muscle mass and function with microgravity duration?s from 30 to 180 days. The results of these studies will have direct benefits to conditions on earth that produce long periods of inactivity, such as, illnesses requiring bed rest, and falls resulting in limb immobilization- the latter is a particularly prevalent problem in the elderly. Of particular importance will be the studies designed to establish the cause of muscle tearing as damage of this sort is a common occurrence in the elderly. Additionally, the results will provide the cellular and molecular data needed to develop effective exercises for slowing the loss of muscle mass that occur with aging.
Task Progress:	In the past year (2004), we completed 90% of the proposed structural and functional studies of the muscle samples from the increment 5 and 6 crew (report filed with NASA). Additionally, we performed pre- and post-flight whole body and cellular studies on the increment 9 crew, pre-flight tests on increment 10 crew, and one year post-flight anslyses on the Russian crew member on increment 7. The latter tests were performed as pre-flight tests were not obtained. The results of the increment 5 and 6 studies were presented at the Integrated Biology of Exercise meeting in Austin, TX on October 6-9, 2004. The results were also published in abstract form in The Physiologist Vol 47, No. 4, August 2004.
Bibliography Type:	Description: (Last Updated: 01/28/2011)
Abstracts for Journals and Proceedings	Fitts, R.H.; Romantowski, J.G.; Lim, W-M; Gallagher, P.; Trappe, S.; Costill, D.; Riley, D.A. "Microgravity, Exercise Countermeasures and Human Single Muscle Fiber Function" Integrated Biology of Exercise Meeting The Physiologist Vol. 47, No. 4, page 320 , Aug-2004
Abstracts for Journals and Proceedings	Riley, D.A; Bain, J.L.; Gallagher, P.; Trappe, S.; Costill, D.; Fitts, R.H. "The Effect of 6-mo Microgravity on Human Skeletal Muscle Structure" Integrated Biology of Exercise Meeting The Physiologist Vol 47, N0. 4, page 320 , Aug-2004
Abstracts for Journals and Proceedings	Gallagher, P.; Trappe, S.; Costill, D.; Riley, D.A.; LeBlanc, A.; Evans, H.; Peters, J.R.; Fitts, R.H. "Human Muscle Volume and Performance: The effect of 6-mo of Microgravity" Integrated Biology of Exercise Meeting The Physiologist Vol 47, N0. 4, page 321 , Aug-2004